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B.Tech - IT
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CSA5112 - CRYPTOGRAPHY AND NETWORK SECURITY

CO1 - ASSESSMENT - 2

APPLICATION - BASED QUESTIONS

1. FIXED SHIFT ENCRYPTION - CAESAR CIPHER :

(a) Encrypt the message using a shift of 4 positions.

Given:

- Plain text : MEET AT NOON
- Shift value : 4

In the Caesar cipher, each alphabet is shifted forward by four positions.

Plain text	M	E	E	T	A	T	N	O	O	N
Shifted letter	Q	I	I	X	E	X	R	S	S	R

Encrypted message : QIIX EX RSSR

Spaces remain unchanged.

(b) How can an authorized member retrieve the original message ?

The authorized member uses the same shift of 4 to decrypt the cipher text.

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Steps:

* Take each encrypted letter.
* Shift it backward by four positions.
* Repeat for every letter.
* Keep the spaces unchanged.

For example:

Q → M
I → E
X → T

Therefore, the decrypted message is MEET AT NOON

Conclusion:

The Caesar cipher provides basic confidentiality by shifting letters according to a fixed key.

2. INTERCEPTED SUSPICIOUS MESSAGE - Atbash Cipher.

(a) Find the original message.

Given cipher text:

GSRH RH Z HVXIVG

The messages uses the Atbash cipher, a substitution cipher in which the alphabet is reversed.

The substitution pairs are:

A ↔ Z, B ↔ Y, C ↔ X, D ↔ W and so on.

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Decrypt each word :

CIPHER TEXT	PLAIN TEXT
GSRH	THIS
RH	IS
Z	A
HVXIVG	SECRET

Original Message : THIS IS A SECRET

(b) Justify your answer.

* The Atbash cipher replaces each letter with its reverse - alphabet counterpart.
* The first word, GSRH becomes THIS
* The second word, RH becomes IS
* The remaining words become A SECRET
* The decoded sentence is meaningful English, confirming the result.

Conclusion:

Atbash is a simple substitution cipher. It is not secure for modern communication because its substitution pattern is fixed and easily recognized.

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3. KEYWORD - BASED ENCRYPTION - VIGENERE CIPHER.

(a). Encrypt HELLO using the keyword KEY.

Given:

Plain text : HELLO
Keyword : KEY

The Vigenere cipher use a repeating keyword to determine the shift for each plain text letter.

Step 1: Repeat the keyword.

Plaintext	H	E	L	L	O
Keyword	K	E	Y	K	E

Step 2 : Convert letters into numerical values.

Use A = 0, B = 1, C = 2, . . . Z = 25

Letter	H	E	L	L	O
Plaintext value	7	4	11	11	14
Keyword letter	K	E	Y	K	E
Keyword value	10	4	24	10	4

Step 3 : Apply the encryption formula.

Formula :

$$C = (P + K) \bmod 26$$

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Where:

C = Ciphertext value

P = Plaintext value

K = Keyword value

Calculations:

- H+K = (7+10) mod 26 = 17 → R
- E+E = (4+4) mod 26 = 8 → I
- L+Y = (11+24) mod 26 = 9 → J
- L+K = (11+10) mod 26 = 21 → V
- O+E = (14+4) mod 26 = 18 → S

Encrypted message: RIJVS.

(b) Explain how the repeating keyword influences the cipher text.

*. Each keyword letter determines the shift applied to the corresponding plaintext letter.

*. The keyword KEY is repeated to match the length of HELLO.

*. Different keyword letter produce different shifts.

*. This makes Vigenere cipher more complex than the Caesar cipher.

*. The receiver must know the same keyword to decrypt the message.

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4. DIGITAL FORENSIC INVESTIGATION - Image Steganography.

(a) How can investigators detect hidden data in an image?

Steganography is a technique used to hide confidential information inside an ordinary file, such as an image, without making the hidden information obvious.

Investigators can use the following methods:

Visual inspection:

Examine the image for unusual colors, patterns or distortions.

File size analysis:

Compare the image's file size with that of similar images to identify unusual differences.

Metadata analysis:

Inspect file metadata for suspicious information or unexpected software details.

Statistical analysis:

Analyze pixel values and color distributions for unusual patterns.

Steganography detection tools:

Use forensic software to identify possible hidden data.

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File extraction:

Examine image data and file structure for appended or embedded information.

Eg: An image may look normal, but its pixel values might contain hidden text.

(b) Recommend two techniques to strengthen the security of hidden communication.

1. Encryption before embedding:

Encrypt the confidential message using a strong encryption algorithm before hiding it inside the image.

- Even if someone extracts the hidden message, they cannot easily understand it without the decryption key.

2. Use randomized embedding locations.

Instead of embedding data in predictable pixel positions, use a secret key to determine where the information is stored.

- This makes the hidden data harder to locate and extract.

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5. TRANSPOSITION - BASED ENCRYPTION :

(a). Find the original message by reversing the encryption process.

Given ciphertext :

WECRLTEERDSOEEFEAOCAIVDEN

* This question refers to a transposition cipher with three levels of arrangement.
* A likely interpretation is the three-rail Rail Fence cipher, a transportation technique that rearranges characters without changing the characters themselves.

The most probable original message is :

WEAREDISCOVEREDFLEEATONCE

However, there is a problem with the question : the supplied ciphertext doesnot match the standard three-rail Rail Fence encryption of this plaintext.

→ For the plaintext

WEAREDISCOVEREDFLEEATONEE,

standard three-rail Rail Fence encryption produces :

WECRLTEERDSOEEFEAOCAIVDEN

So, the ciphertext can be reversed to recover the original message :

WEAREDISCOVEREDFLEEATONCE

With space inserted :

WE ARE DISCOVERED FLEE AT ONCE.

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(b) Why is this method vulnerable in modern communication systems ?

No character substitution:

The original letters remain unchanged; only their positions are rearranged.

Limited key space:

The no. of possible arrangements is small for simple transposition methods.

Frequency patterns remain:

Letter frequencies are preserved, which can help attackers analyze the cipher text.

Easy to break:

Attackers can try different arrangements to recover the original message.

No modern security protection:

A basic transposition cipher does not provide strong protection against modern cryptanalysis.

(c). Suggest one practical enhancement.

Combine transposition with strong encryption.

Encrypt the plaintext using a modern, authenticated encryption algorithm, such as AES-GCM, rather than relying on a simple transposition cipher alone.

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